Response Letter with Annotations

Muddiman, A., McGregor, S.C., Stroud, N.J. (2018). (Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries. *Political Communication,* 36(2), pp. 214-226. https://doi.org/10.1080/10584609.2018.1517843.

Annotations

1. **Purposes**
    a. **Motivation (one sentence summary)** – The motivation of this study was to develop and share a new methodology for computational content analysis – manually validated organic dictionaries – through a study on incivility in online news comments.
    b. **Specific Rationale (several bullet points)**
        i. To allow for context-dependent language evaluation
        ii. More nuanced computational analysis of content
    c. **Research Questions and/or Hypotheses – Identify IV, DV, mediator, moderator/ Specific relationships among variables**
        i. This is a primarily methodological paper rather than a traditional hypothesis-testing study, and so the main focus of this paper was to introduce and validate a new computational methods
        ii. The variables are the different measurement approaches: (i.e. manually validated organic dictionaries (their proposed method), sentiment dictionaries, human coding, and API's (machine learning)).
        iii. The outcome being measured is the accuracy in identifying context-dependent language. The paper is about measurement validity.
2. **Methods**
    a. **Research Design**
        i. (1) Created a list of the top words in existing data set of news comments on New York Times website
        ii. (2) Removed stop words (e.g. "the," "a," etc.)
        iii. (3) used Porter (1980) stemmer to remove the endings of words and combine word roots (i.e. mislead/misleading)
        iv. (4) Ran National Language Toolkit program (Bird et al., 2009) in Python to compile 5,000 features used most often in data set
        v. (5) Deductive approach to selecting features related to focus of dictionary (incivility) – overselection of potential related features
        vi. (6) Human coder validation of each feature – 25 comments for each of the 88 potential incivility features
        vii. (7) created separate dictionary of swear words from the *Times* data set to compare results to swear words (See Table 1)

viii. (8) Tested a sample of 400 comments against existing methods (i.e. stratified random sampling; LIWC; Lexicoder, etc. See Table 2)
ix. (9) Replication: second study designed to compare the language used in the *Chattanooga Times Free Press* Pulitzer finalist poverty feature series with language used in tweets.
x. (10) Assessed how many of the more than 5 million geotagged tweets in the original data set discussed poverty using mirroring language from *Times Free Press* coverage.

b. **Sample (representative sample?)**
   i. Original Study: Existing data set from comments on New York Times Article
   ii. Validation: testing 400 comments against existing methods
   iii. Replication:5 million geotagged tweets from dataset

c. **Measures/Analysis**
   i. Deductive approach – Created dictionary with features from existing dataset
   ii. Human coder validation
   iii. Replication with separate data set

3. **Results and Conclusions**

   a. **Answers to RQs and Hypotheses** – This article looked to describe a methodological approach and did not include traditional RQs or hypotheses, but instead provided recommendations for future methodological implementation.
      i. (1) There were favorable outcomes for manually creating dictionaries from content to account for context of the content. Through the manual validation of the specific word use, the organic dictionary offers research the ability to capture terms that may have varying or nuanced meanings that are context-dependent, rather than working with a standardized dictionary system that may not conceptually align with the research topic.
      ii. (2) Unable to capture content in totality, but this approach was favorable compared to sentiment analysis, Google Perspective API, and trained human coding.
      iii. (3) Manual dictionaries contained less words than more standardized, out-of-the-box dictionaries, but the manual dictionary was more aligned with the concept being measured. “It is important to be precise and confident that the terms in the dictionary match the concept” (p. 223).

   b. **Include Key Figures/Tables**

i.

Table 1 Dictionaries

New York Times Study		Chattanooga *Times Free Press* Study
Swear Words	**Uncivil Words**	**Poverty**
Included: ass, bs, crap, damn, hell	Included: dumb, farc, hypocrit, insan, insecur, irrespons, sham, trivial, unaccept, uneth, unfair, bigot, bigotri, discrimin, prejudic, racism, racist, segreg, stereotyp, betray, enemi, insurg, overthrow, riot, threat, threaten, traitor, treason, tyranni, brainwash, deceiv, decept, dishonest, disingenu, inaccur, incorrect, misinform, mislead, uninform, dysfunct, infring, obstruct, obstructionist, suppress, unconstitut, chao, debacl, delud, demean, disrespect, fiasco, inappropri, nasti, vitriol, yell	Included: employment, good-paying, income, inequality, low-income, low-wage, poverty, recession, unemployment, upward-mobility, wage
Not Included: screw, suck, swear	Not included: cheat, controversi, crimin, desper, devil, dismal, egregi, helpless, horrend, horribl, horrif, illeg, miseri, neglig, squander, terribl, unpleas, offend, endang, lynch, revolt, disenfranchis, filibust, impeach, bogu, distort, fact, irrelev, lie, brutal, meltdown, outrag, toxic	Not included: activist, Appalachian, at-risk, benefit, blue-collar, career, development, economic, economy, employee, employer, equality, expense, good-paying, hardworking, higher-wage, homeless, income, inequality, lower-income, low-income, low-wage, middle-class, mobility, poverty, problem-solve, prosperity, puzzle, recessions, reform, solution, struggling, support, underclass, unemployment, upper-class, wage, wealth, welfare, white-collar, working-class, abandon, afford, challenge, circumstance, class, community, communities, deficit, develop, disadvantaged, disconnected, earnings, economically, economist, employed, employment, finance, financial, hardship, isolated, job, opportunities, oppressed, poor, prosperous, recession, redevelopment, resources, rich, salary, shelter, stability, stabilized, struggle, upward-mobility, work, workforce

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ii.

Table 2 Comparison of Methods to Gold Standard

	% Correct	% False Positive	% False Negative
Manually validated organic dictionaries	73.1	4.3	22.8
Lexicoder	68.3	30.8	1.0
LIWC	70.8	22.8	6.5
Toxicity	74.6	8.8	16.8
Trained coder #1	70.1	1.3	28.7
Trained coder #2	83.3	10.3	6.5

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c. **Author's Conclusions**

  i. Large scale, context-dependent content analysis is feasible through the manual creation of dictionaries and the manual validation of each feature. This can "help scholars balance the need for theoretical control over computer categorization of content and the need for flexible analyses that take context into account" (p. 224).

4. **Discussion**

a. **Implications** – Major methodological contribution in establishing a way to accurately perform large-scale context-dependent content analysis through computational means.

b. **Limitations**

  i. Incomplete capture of content
  ii. Temporal specificity of dictionaries
  iii. Comparisons are imprecise because of the nature of the different tools used

c. **Future Directions**

  i. Authors suggested to empirically test the optimal size of features list, the appropriate number of texts to sample per feature, and the best threshold percentage for feature inclusion
  ii. Combination with other methods, and applications to different concepts

Response Letter

In this article, Muddiman et al. (2019) address a critical hole in computational text analysis methodology: how to analyze large-scale datasets when studying context-dependent concepts. The authors introduced manually validated organic dictionaries as a hybrid approach to combine computational efficiency with theoretical precision in the analysis of these large datasets. In attempting to identify political incivility in over 9 million *New York Times* comments, researchers faced difficulty in coding context-dependent language on which much of the terms being parsed relied.

To address this problem, researchers first created a list of the most frequently used word stems, or features, in the dataset, and then selected features that were potentially relevant to the concept based on previous theoretical research. Then, the researchers manually validated each feature by having trained coders review 25 randomly selected texts containing that feature. Features were only included in the final dictionary when 80% or more of the sampled texts used the feature in alignment with the context. Ultimately, this process reduced their initial potential incivility features to a more manageable number (55 validated features). These features appeared in 10% of the comments.

This article also provides insight into the validation process of this methodology. The authors validated the approach by creating a swear words dictionary that closely matches established taboo word lists, that would indicate that the word is colloquially considered a swear word. The authors also compared the method against sentiment dictionaries (LIWC and Lexicoder), Google's Perspective API, and human coders. The manually validated dictionaries achieved more than 70% accuracy and less than 5% of a false-positive rate. This method thus outperformed sentiment dictionaries and at least matched the accuracy of other methods. The researchers also replicated the approach in a second study examining poverty-related language in tweets from Chattanooga, Tennessee, and also received favorable results.

While not all concepts translate well to automated analysis, the researchers strike a nice balance of leveraging theoretical expertise while acknowledging the benefits of large-scale analysis. In addition, the study was transparent and replicable. Training data for computational tools can often be unknown or opaque (such as with Perspective AI), but the manually validated dictionaries allow other researchers to understand what is being measured and to reproduce the analysis in other contexts.

The greatest strength of this study is the ability to balance scalability with validity, which I would argue is the study's main contribution to computational methods in mass communication literature. However, the study had notable limitations. It cannot capture all the relevant content, and so there will always be some holes in the data. Additionally, dictionaries that are manually created in this way are context-specific and temporal, and thus other researchers may not be able to pull from the dictionaries themselves, and new dictionaries will likely have to be made and updated regularly as world events continue and circumstances change, and new topics arise in discourse online.